Phase 1: Project Declaration

Group Assignment: Noise Schedule Ablation in Diffusion Models

Course: Foundation Models and Generative AI | Submission Deadline: October 10, 2026

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1. Chosen Image-to-Image Task

Task Definition: Semantic Facade Label Map to Photorealistic Building Facade Synthesis.

The objective is structural image-to-image translation where the diffusion model takes a 2D discrete semantic layout as conditioning input and synthesizes a rectified photorealistic photograph of the corresponding building facade. The input layout consists of piecewise-constant color regions marking architectural components (walls, windows, doors, sills, moldings). The synthesized output must preserve the boundaries and geometric structure specified by the input segmentation map while rendering realistic masonry textures, glass reflections, and natural illumination.

Dataset Controls and Preprocessing: Because CMP images possess varying native aspect ratios, images will be standardized to 256×256 resolution. Semantic label maps will be resized strictly via nearest-neighbor interpolation using a fixed color palette to prevent synthetic class blending at object boundaries. The dataset will be partitioned using a fixed random seed into 400 training, 100 validation, and 106 test images. Inference and training across all runs will condition on the identical text prompt: "turn this semantic facade label map into a photorealistic building facade".

2. Chosen Pretrained Model and Codebase

- **Model Architecture:** InstructPix2Pix (Brooks, Holynski, Efros, CVPR 2023, arXiv:2211.09800), official codebase at https://github.com/timothybrooks/instruct-pix2pix and weights checkpoint at timbrooks/instruct-pix2pix.
- **Conditioning Architecture:** Adapts the Stable Diffusion UNet by adding 4 extra input channels to the first convolutional layer, concatenating the VAE latent of the conditioning image E(c_I) directly with noisy latent z_t (total 8 input channels), plus text cross-attention conditioning c_T.

3. Chosen Dataset and Source URL

- **Dataset Name:** CMP Facade Database (Center for Machine Perception, Czech Technical University in Prague), URL: https://cmp.felk.cvut.cz/~tylecr1/facade/.
- **Dataset Details:** Contains 606 rectified facade photographs (378 base, 228 extended) across diverse cities around the world and various architectural styles, paired with 12-class manually annotated segmentation masks (facade, molding, cornice, pillar, window, door, sill, blind, balcony, shop, deco, background).

4. Evidence that the Dataset Was Not Used During Pretraining (Option b)

Under Option (b) of the assignment guidelines, we cite the pretraining sources for both stages of the model architecture and demonstrate why the paired label-map task and conditioning modality are outside pretraining. (Note: For arXiv:2211.09800v2, citations reference the camera-ready two-column layout with line numbers counted sequentially from the top of the designated column; line numbers for the model card reference the raw commit permalink at https://huggingface.co/stable-diffusion-v1-5/stable-diffusion-v1-5/raw/451f4fe16113bff5a5d2269ed5ad43b0592e9a14/README.md).

- **Base Model Architecture and Pretraining (Stable Diffusion v1.5):** The Latent Diffusion Model architecture is described in Rombach et al., "High-Resolution Image Synthesis with Latent Diffusion Models", CVPR 2022 (arXiv:2112.10752), Section 3 (pp. 3–5, specifically Sec. 3.2–3.3 on pp. 4–5 detailing latent diffusion and conditioning). Pretraining data lineage of the v1.5 weights is documented in the pinned official model card (stable-diffusion-v1-5/stable-diffusion-v1-5, commit 451f4fe, lines 153–160): v1.1 was trained on laion2B-en (237k steps at 256×256) and laion-high-resolution (194k steps at 512×512), resumed as v1.2 on laion-improved-aesthetics (515k steps at 512×512), and resumed as v1.5 on laion-aesthetics v2 5+ (595k steps at 512×512).
- **Instruction Tuning Stage (InstructPix2Pix):** Brooks et al., "InstructPix2Pix: Learning to Follow Image Editing Instructions", CVPR 2023 (arXiv:2211.09800v2): Section 3.1.1 (p. 3, col. 2, paras. 2–3, lines 1–28, describing GPT-3 generation of 454,445 text edit instructions and paired captions from LAION-Aesthetics), Section 3.1.2 (p. 3, col. 2, lines 29–36; p. 4, col. 1, lines 22–45; p. 4, col. 2, lines 9–16, detailing Prompt-to-Prompt paired image synthesis across $p \in [0.1, 0.9]$ and CLIP filtering), Section 3.2 (p. 4, col. 2, lines 17–43; p. 5, col. 1, lines 12–21, detailing conditional diffusion training initialized from a pretrained Stable Diffusion

checkpoint, with additional input channels concatenating z_t and E(c_I)), and Appendix A.3 (p. 14, “Training InstructPix2Pix” paragraph, stating initialization from the Stable Diffusion v1.5 EMA checkpoint and training at 256×256 resolution).

Domain Separation Argument:
InstructPix2Pix is initialized from Stable Diffusion v1.5, whose pretraining data is a web-scraped LAION subset of natural image–caption pairs [Rombach et al., CVPR 2022, Sec. 3, pp. 3–5; SD v1.5 Model Card, lines 153–160], and is then fine-tuned on synthetic paired examples generated from 454,445 GPT-3 edit instructions and Prompt-to-Prompt image pairs [Brooks et al., CVPR 2023, Sec. 3.1–3.2, pp. 3–5, and App. A.3, p. 14, as cited above]. Neither stage's documentation describes paired semantic segmentation label maps and photographs, and neither documents use of the CMP Facade Database. The CMP conditioning inputs are piecewise-constant 12-class label maps, an entirely different input modality from natural photographs paired with text edits. The model developers explicitly note that no dataset deduplication was performed during web pretraining (model card, line 116), so photographic facade overlap in web-scale LAION data cannot be formally ruled out; our argument is therefore that the paired label-map-to-photo task and input conditioning modality are absent from pretraining, not that no facade photo was ever seen. As a result, Run 0 baseline failure is expected (as a primary hypothesis) from the unseen input modality, providing a clear test of out-of-distribution fine-tuning.

5. Native Noise Parameterization of the Chosen Model

InstructPix2Pix uses the discrete Gaussian diffusion forward process of Stable Diffusion over T = 1000 timesteps. The native noise parameter of the model is the per-step variance:

$$\theta_t = \beta_t$$

Proof of Boundedness ($\theta_t \in (0, 1)\ \forall\ t$): Under the native scaled linear schedule of Stable Diffusion, the variance parameter is defined as $\beta_t = (\sqrt{\beta}_{start} + (t - 1)/(T - 1) * (\sqrt{\beta}_{end} - \sqrt{\beta}_{start}))^2$, where $\beta_{start} = 0.00085$ and $\beta_{end} = 0.0120$. Over all timesteps $t \in \{1, 2, ..., 1000\}$, β_t increases monotonically from $\beta_1 = 0.00085$ to $\beta_{1000} = 0.0120$. Because $0.00085 \le \beta_t \le 0.0120 < 1$ for all t, the condition $\theta_t \in (0, 1)\ \forall\ t$ is strictly satisfied. Associated quantities are $\alpha_t = 1 - \beta_t \in (0, 1)$ and the cumulative retention factor $\alpha_{bar,t} = \prod_{s=1}^{t} \alpha_s \in (0, 1)$.

6. Chosen Exponent x and Schedule Analysis

We declare the power exponent x = 2.20, which lies strictly within the required interval (1.5, 3.0). The ablated schedule for Run B is defined elementwise as $\{\theta_t^x\}_{t=1}^{1000} = \{\beta_t^{2.20}\}_{t=1}^{1000}$.

Because $\beta_t \in (0, 1)$, applying an exponent $x > 1$ strictly shrinks every variance value: $\beta_t^{2.20} < \beta_t$. Specifically, the modified endpoints evaluate to $\beta_1^{2.20} = (0.00085)^{2.20} \approx 1.76 \times 10^{-7}$ and $\beta_{1000}^{2.20} = (0.0120)^{2.20} \approx 5.95 \times 10^{-5}$. The table below traces the cumulative signal retention multiplier $\alpha_{bar,t}$ and the resulting Signal-to-Noise Ratio $SNR(t) = \alpha_{bar,t} / (1 - \alpha_{bar,t})$ across five representative diffusion timesteps:

t	$\alpha_{bar,t}$ (Native)	$\alpha_{bar,t'}$ (x=2.20)	SNR Native	SNR Modified
1	0.99915	0.9999998	1,175	5.7×10^6
250	0.675	0.99981	2.08	5,322
500	0.278	0.99874	0.38	795
750	0.057	0.99500	0.060	199
1000	0.0047	**0.985**	0.0047	**66**

Mechanistic Implications for Phase 2: In the original schedule, $\alpha_{bar,1000} \approx 0.0047$, so the latent collapses almost completely to isotropic Gaussian noise. In contrast, under the ablated schedule, $\alpha_{bar,1000'} \approx 0.985$, meaning the forward process retains ~98.5% of the original signal variance (with signal amplitude multiplier $\sqrt{\alpha}_{bar,1000'} \approx 0.9925$, resulting in an elevated SNR ≈ 66) and never approaches pure noise. In our experimental design, reverse inference in both runs will strictly initiate from standard Gaussian noise $x_T \sim N(0, I)$. Because the Run B model never encounters near-pure noise during fine-tuning, this creates an acute distributional mismatch at t = T during reverse sampling. Furthermore, Run B fine-tunes from checkpoint weights whose initial representations were learned under native variance levels, adding a secondary inductive mismatch. We hypothesize that Run B will show degraded sample quality compared to Run A, providing an empirical test of the schedule mechanics. In implementation, all schedule quantities and sampler sigmas will be precomputed in float64 before casting to prevent precision loss near 1, and live scheduler buffers will be verified against the analytical table.

7. Planned Experimental Runs for Phase 2

Run	Condition	Schedule	Configuration and Purpose
Run 0	Pre-Fine-Tuning Baseline	Native {β_t}	Apply pretrained InstructPix2Pix directly to CMP Facade test inputs without fine-tuning. Tests zero-shot behavior on the unseen label-map input modality.
Run A	Fine-Tuning (Original)	Native {β_t}	Fine-tune full UNet on CMP Facade for 20 epochs (100 steps/epoch at batch size 4, total 2000 steps; learning rate 1e-5, AdamW weight decay 1e-2, FP16 mixed precision) using native schedule. Checkpoints saved every 4 epochs (5 checkpoints).
Run B	Fine-Tuning (Ablated)	Modified {β_t^2.20}	Fine-tune under identical hyperparameters, seed, optimizer, and epochs using the power-transformed schedule {β_t^2.20}.

Phase 2 Sampler Consistency Note: All inference runs (Run 0, Run A, Run B) will utilize strictly identical sampling parameters: 50 inference steps, fixed random seed 42, text guidance scale s_T = 7.5, and image guidance scale s_I = 1.5.

8. Group Declaration

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